

MECH 191 — Metallurgy

Week 5

Stainless Steels & Cast Irons

September 21, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Stainless Steels

What makes it stainless, and why the grade matters as much as the material family

02

Stainless Families

Austenitic, ferritic, martensitic, duplex, and PH — microstructure drives every property difference

03

Cast Irons

Why cast iron is not steel, and how the form of excess carbon determines the type

Module 4 · Week 5

Stainless Steel & Cast Irons

What to do this week:

Distinguish graphite flakes (E) from nodules (F). Find annealing twins in G at 100×

Steps:

- 1 Observe — complete feature table
- 2 Sketch — label magnification & specimen
- 3 Answer — 4 short questions

Specimens to Check Out

- Specimen E — gray cast iron
 - Specimen F — ductile cast iron
 - Specimen G — 304 SS
-

How to check out the kit:

- 1 See instructor after class today
- 2 Sign out the specimen tray
- 3 Use the EXAMET-5 at your convenience
- 4 Return kit before leaving class
- 5 Submit completed module with your portfolio

Portfolio Handout — Module 4

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 ← here Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 5 — At a Glance

Topics

- What makes stainless steel corrosion-resistant — the passive chromium oxide layer
- The four stainless steel families and how microstructure defines each one
- Austenitic (300s): FCC, Ni-stabilized, non-magnetic, best corrosion resistance
- Ferritic & martensitic (400s): BCC/BCT, magnetic, hardenable grades
- Cast iron as a distinct iron-carbon family — not a low-quality steel
- Gray, white, ductile, and malleable cast iron — how carbon form drives properties
- When and why to choose cast iron over steel

Resources

Reference Material

Moniz 5th — Ch. 16 & 17
Cast Irons / Stainless Steels

Kalpakjian 6th — Ch. 6
Ferrous Metals and Alloys

Visualizer

Iron-Carbon Phase Diagram
(Interactive)

Reference

AISI/SAE Steel Designation Codes

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- 1 What single threshold separates stainless steel from ordinary alloy steel — and why does that exact percentage matter?
- 2 A coastal facility needs a tank for chloride brine: 304 or 316? What is the deciding factor?
- 3 Two castings from the same iron-carbon melt: one becomes gray iron, one white iron. What determined which was which?
- 4 Why is gray cast iron standard for engine blocks and machine bases, despite having low tensile strength?
- 5 A martensitic stainless weld cracks in the heat-affected zone. What most likely went wrong, and how do you prevent it?

What Makes Stainless Steel 'Stainless'?

All stainless steels share one feature: a minimum of 10.5% chromium by weight. That single threshold is what separates stainless from ordinary alloy steel.

The Passive Chromium Oxide Layer

- Chromium reacts with atmospheric oxygen to form a thin, stable Cr_2O_3 layer on the surface — invisible, only a few nanometers thick
- This passive layer acts as a barrier, preventing oxygen and moisture from reaching the steel beneath
- Self-repairing: if scratched or abraded, the layer reforms spontaneously in the presence of oxygen
- Destroy the passive layer (with chlorides, oxygen-poor environments, or mechanical damage in an aggressive medium) and corrosion can proceed rapidly
- 10.5% Cr is the threshold — below this, not enough Cr_2O_3 forms to maintain a complete, stable passive layer

Key Alloying Thresholds

≥ 10.5% Cr

Stainless threshold

Passive layer forms. Now classified as stainless steel.

+ Nickel

Austenite stabilizer

Keeps FCC structure at room temp → non-magnetic, ductile, tough.

+ Molybdenum

Pitting resistance

Improves resistance to chloride pitting (e.g., 316 vs. 304).

+ Carbon

Hardenability

Enables hardening by heat treatment → martensitic grades.

Stainless Steel Families — Microstructure Is Everything

Same passive chromium oxide layer — very different microstructures. The balance of alloying elements determines which phase is stable at room temperature, and that phase determines everything.

Austenitic

300 series

FCC

Non-magnetic · Cold work only
Cr: 18–20% / Ni: 8–12%

Nickel stabilizes the FCC austenite phase all the way to room temperature. Most widely used family — best combination of corrosion resistance, formability, and weldability. Cannot be hardened by heat treatment.

304 (general purpose) · 316 (+ Mo for chloride resistance)

Food equipment, chemical vessels, marine hardware, kitchen appliances, medical devices

Ferritic

400 series

BCC

Magnetic · Not hardenable
Cr: 10.5–30% / Ni: No Ni

High chromium content stabilizes the BCC ferrite structure. No nickel makes these grades less expensive. Moderate corrosion resistance, lower toughness in thick sections and at low temperatures.

430 (most common) · 439 · 446

Automotive trim and exhausts, kitchen appliances, architectural panels, dishwasher liners

Martensitic

400 series

BCC/BCT

Magnetic · Quench & temper
Cr: 10.5–18% / Ni: Low or none

Higher carbon content allows martensite to form on quenching. Can be hardened to high strength — but at the cost of lower corrosion resistance and toughness compared to austenitic grades. Requires preheat and PWHT to weld.

410 · 420 · 440C

Cutlery, surgical instruments, valve stems, pump shafts, turbine blades

Duplex

Various

Austenite

+ Ferrite

Semi-magnetic · Not by HT
Cr: 21–26% / Ni: 4–8%

Mixed austenite/ferrite microstructure — higher strength than austenitic, better resistance to stress corrosion cracking (SCC). Increasingly common in oil & gas, chemical processing, and marine environments.

2205 · 2507

Subsea pipework, chemical plant, heat exchangers, desalination equipment

Austenitic Stainless — The Workhorse Family (300 Series)

Austenitic grades account for over 70% of all stainless steel produced. 304 and 316 are the two grades a technician will encounter most often.

304

General-Purpose Austenitic

18% Cr / 8% Ni — the most widely used stainless steel in the world

- Excellent corrosion resistance in most atmospheric, food, and chemical environments
- FCC structure — very ductile and tough, excellent formability
- Non-magnetic in annealed condition (can become slightly magnetic after cold work)
- Good weldability — low carbon 304L grade available for thicker weld sections to prevent sensitization
- Cannot be hardened by heat treatment — only by cold work
- Susceptible to chloride pitting and stress corrosion cracking (SCC) in chloride environments

Uses: sinks, countertops, food processing, chemical tanks, medical trays, architectural panels

316

Marine & Chemical Grade

18% Cr / 10% Ni / 2-3% Mo — the upgrade for chloride environments

- Molybdenum addition dramatically improves resistance to chloride pitting and crevice corrosion
- The grade of choice for marine environments, seawater, and halide-containing process streams
- Same FCC structure and non-magnetic behavior as 304
- 316L (low carbon) is preferred for welded assemblies to prevent sensitization
- Higher cost than 304 — specify it when 304 has failed in service or chlorides are known to be present
- Used in pharmaceutical and food processing where frequent washdowns with chlorine-based cleaners occur

Uses: marine fittings, chemical plant, pharmaceutical vessels, coastal architecture, lab equipment

Ferritic & Martensitic Stainless — The 400 Series

Both are magnetic BCC-based grades. Ferritic cannot be hardened; martensitic can — but both trade corrosion resistance for other performance priorities.

Ferritic (e.g. 430)

BCC structure • Magnetic • No nickel • NOT hardenable by heat treatment

- BCC ferrite phase at all temperatures — high chromium (10.5–30%) stabilizes BCC and prevents transformation to austenite
- No nickel = lower cost than austenitic grades
- Moderate corrosion resistance — good for mild environments, not suitable for chloride-rich or highly aggressive conditions
- Limited toughness, especially at low temperatures and in thick sections
- Good oxidation resistance at elevated temperatures — useful in exhaust systems
- Magnetic — can be detected and separated with magnets (useful in recycling/fabrication)

Martensitic (e.g. 410, 420)

BCC/BCT structure • Magnetic • Hardenable • Higher carbon than ferritic

- Higher carbon content allows martensite to form when austenite is quenched — the same mechanism as tool steels, applied to a stainless grade
- Can be hardened to high strength and hardness by quench and temper — the only stainless family with significant heat treatment response
- Lower corrosion resistance than austenitic or ferritic grades (less Cr, more C)
- Magnetic in all conditions
- Requires preheat (typically 200–300°C) and post-weld heat treatment when welding — failure to do so causes hard, brittle heat-affected zones prone to cracking
- 410 (general purpose) • 420 (higher C, cutlery) • 440C (maximum hardness)

Welding Note:

Ferritic: limited weldability in thick sections — grain growth reduces toughness. Martensitic: **MUST** preheat; PWHT strongly recommended. Never weld martensitic in the hardened condition.

Cast Iron — A Distinct Family, Not a Low-Quality Steel

Cast iron is NOT steel that didn't make it. It is a separate family of iron-carbon alloys with its own unique microstructures, properties, and engineering roles.

What Defines Cast Iron

- Carbon content above 2.14% — typically 2.5–4% plus significant silicon (1–3%)
- At these carbon levels, the excess carbon cannot stay in solid solution — it must precipitate out during solidification
- The FORM the carbon takes when it precipitates determines everything about the cast iron type
- Silicon content affects how carbon precipitates: more Si → graphite forms; less Si → iron carbide (cementite) forms
- Cast iron is poured into complex shapes — generally not wrought, rolled, or drawn into structural sections
- Melting point lower than steel (~1150–1200°C vs. ~1500°C) — easier to pour into complex molds
- Generally more brittle than steel — most grades cannot absorb significant impact loading

Steel vs. Cast Iron — Key Differences

Property	Steel	Cast Iron
Carbon	0.02–2.14%	> 2.14% (typ. 2.5–4%)
Structure	Wrought (rolled, forged, drawn)	Cast (poured into molds)
Ductility	Good to excellent	Poor (most grades)
Toughness	Good — absorbs impact	Poor — brittle fracture
Machinability	Moderate	Excellent (gray iron esp.)
Damping	Low	Excellent (gray iron)
Weldability	Fair to good	Poor — very difficult
Cost/shape	Higher for complex shapes	Cheap for complex shapes

Gray & White Cast Iron — Two Extremes

The same iron-carbon composition can produce dramatically different materials depending on cooling rate and silicon content — which determines the form carbon takes.

Gray Cast Iron

Carbon precipitates as GRAPHITE FLAKES — slow cooling, higher silicon

- Graphite flakes act as internal stress concentrators — crack initiation sites that limit tensile strength
- Poor tensile strength but excellent compressive strength — used where loads are compressive
- Graphite provides self-lubrication and acts as a chip breaker — reduces wear and aids machining
- Outstanding vibration damping — graphite absorbs energy; ideal for machine bases and engine blocks
- Fracture surface appears gray (hence the name) — identified visually
- Inexpensive and easy to cast into complex shapes — most widely produced cast iron

Engine blocks, machine tool bases, brake rotors, pipe, manhole covers, cookware

White Cast Iron

Carbon locked as IRON CARBIDE (Fe_3C / cementite) — rapid cooling, lower silicon

- Rapid cooling (or low Si content) prevents graphite from forming — carbon stays as cementite (Fe_3C)
- Cementite is extremely hard and brittle — Rockwell C 65–70+ in some compositions
- Very high wear resistance — resists abrasive wear exceptionally well
- Not machinable by conventional cutting tools — must be ground
- Very brittle — fractures with little or no warning, cannot absorb impact
- Fracture surface appears white or light-colored (cementite is pale)
- Starting material for malleable cast iron — white iron is annealed to convert cementite to carbon rosettes

Mill liners, crusher plates, grinding media, pump wear parts, starting stock for malleable iron

Ductile & Malleable Cast Iron — Engineering-Grade Cast Irons

Both achieve improved ductility by changing the shape of the carbon phase — eliminating the sharp flakes that make gray iron brittle.

Ductile (Nodular) Iron

Magnesium or cerium added to the melt just before casting — graphite forms as SPHEROIDS, not flakes

- Spheroidal graphite shape eliminates the notch effect of flakes — no internal stress concentrators
- Dramatically improved ductility, toughness, and tensile strength compared to gray iron
- Tensile strength 400–900 MPa — approaching medium-carbon steel performance
- Can be heat treated to further adjust properties
- Good castability — complex shapes achievable at lower cost than steel castings
- The most widely used cast iron for structural and load-bearing applications

Crankshafts, gears, heavy-duty pipe, automotive suspension, wind turbine housings, pump bodies

Malleable Cast Iron

White cast iron annealed at high temperature — cementite converts to rounded TEMPER CARBON rosettes

- Two-step process: cast as white iron (rapid cooling), then anneal for many hours at ~900°C to convert cementite
- Temper carbon rosettes have rounded edges — much less severe stress concentration than graphite flakes
- Good ductility and machinability — significantly better than white or gray iron
- Slower, more expensive process than producing ductile iron — largely being replaced in modern applications
- Still used for thin-section castings where ductile iron has difficulty (small fittings, hand tools)

Pipe fittings, hand tools, brackets, agricultural equipment, automotive parts (legacy)

Cast Iron — Choosing the Right Grade

Cast iron is chosen when its specific characteristics are an advantage, not a limitation. Understanding which property each type excels at is the key to correct grade selection.

Gray Iron

Graphite
Flakes

Strength: Low tensile
high compressive
Damping: Excellent
Machin.: Excellent
Toughness: Poor

Choose when:

Best vibration damping of any common engineering metal. Choose for machine bases, engine blocks, brake rotors, and anything where vibration absorption matters.

Avoid when:

Any application with significant impact or tensile loading. Avoid where fracture toughness is needed.

White Iron

Iron Carbide
(Fe₃C)

Strength: Very high comp.
no tension
Damping: Poor
Machin.: Not machinable
Toughness: Very poor

Choose when:

Maximum wear resistance is the priority and brittleness can be tolerated. Ideal for mill liners, crusher plates, and grinding equipment.

Avoid when:

Structural applications. Any impact loading. Cannot be welded or machined conventionally.

Ductile Iron

Graphite
Spheroids

Strength: High — up to
900 MPa
Damping: Moderate
Machin.: Good
Toughness: Good

Choose when:

Need good mechanical properties at lower cost than steel castings. Crankshafts, gears, heavy pipe, and structural castings where ductile iron replaces steel forgings.

Avoid when:

Extreme impact or shock loading — still less tough than steel. Not suitable where welding is required.

Malleable Iron

Temper Carbon
Rosettes

Strength: Moderate
Damping: Moderate
Machin.: Excellent
Toughness: Moderate

Choose when:

Thin-section castings requiring ductility and machinability where ductile iron is difficult to cast. Fittings, hand tools, and small hardware.

Avoid when:

Thick sections — process economics make ductile iron more attractive. Being phased out for most new applications.

Stainless & Cast Iron — Fabrication Considerations

Stainless steels and cast irons each present specific challenges in the shop. Understanding why these requirements exist is more valuable than just following a checklist.

1

Sensitization in Austenitic Stainless

When austenitic stainless is held at 425–815°C (during welding), chromium carbides precipitate at grain boundaries — depleting nearby zones of Cr below 10.5%. This 'sensitized' zone is no longer corrosion-resistant. Solution: use low-carbon grades (304L, 316L) or stabilized grades (321, 347 with Ti/Nb). Always specify L-grade for welded fabrication in corrosive service.

2

Martensitic Stainless — Preheat & PWHT

Martensitic stainless must be preheated (typically 200–300°C) before welding to slow cooling and prevent martensite formation in the heat-affected zone before hydrogen can escape. PWHT (temper at 650–750°C) restores toughness. Welding in the hardened condition without these steps causes cracking — the HAZ becomes a brittle fracture initiation site.

3

Cast Iron Welding — Why It's Difficult

High carbon content means high carbon equivalent — virtually all cast irons have CE > 1.0, requiring extreme preheat (400–600°C for gray iron) or cold welding with nickel-based filler. The primary risk is white iron formation in the HAZ as carbon migrates and cementite forms during rapid cooling. Pre-heat slow-cool and stress relief are mandatory for structural welds in gray and ductile iron.

4

Stainless Contamination Control

Stainless steel passive layers are destroyed by contact with carbon steel particles (rust transfer), chloride-containing cleaners, and abrasive contamination. In fabrication: use dedicated stainless tools, discs, and wire brushes. Never use carbon steel wire brushes on stainless. Pickle and passivate welds and heat-affected zones on austenitic stainless after fabrication to restore full corrosion resistance.

Core Questions — Answers

1

What threshold makes stainless 'stainless,' and why that exact value?

≥ 10.5% Cr. Below this, not enough Cr_2O_3 forms to maintain a complete passive layer. Above this, the layer forms spontaneously and self-repairs when scratched.

2

304 or 316 for a chloride-brine tank — and why?

316. The 2–3% Mo addition dramatically improves resistance to chloride pitting and crevice corrosion — exactly the failure mode seawater causes in 304.

3

Same melt, two castings: one gray iron, one white iron. What determined which?

Cooling rate and silicon content. Slow cooling + higher Si → graphite flakes form (gray iron). Rapid cooling + lower Si → carbon stays as cementite/ Fe_3C (white iron).

4

Why gray cast iron for engine blocks despite low tensile strength?

Graphite flakes provide outstanding vibration damping and excellent compressive strength — the loads are compressive. Graphite also self-lubricates and aids machining.

5

Martensitic stainless HAZ cracks after welding. What went wrong?

Missing preheat (200–300°C) and/or PWHT. Without preheat, hydrogen is trapped in brittle martensite. Without PWHT (temper 650–750°C), the HAZ remains hard and fractures.

Key Takeaways

1 Stainless steel is a family — the grade determines the microstructure

Specifying 'stainless' without a grade is insufficient. The microstructure (FCC austenite, BCC ferrite, BCT martensite) determines magnetic behavior, hardness response, toughness, and corrosion resistance in each service environment.

2 10.5% chromium creates the passive layer — everything else is fine-tuning

Ni stabilizes austenite. Mo improves chloride resistance. C enables hardening. Si and Cr content determine which phase is stable. Every grade decision traces back to what the passive layer needs to survive in the intended environment.

3 Cast iron's properties are determined by the form its excess carbon takes

Graphite flakes (gray) → damping and machinability but brittle. Spheroids (ductile) → strength and toughness. Carbides (white) → extreme hardness but very brittle. Temper carbon (malleable) → improved ductility after annealing.

4 Cast iron is not failed steel — it is a deliberate engineering choice

When damping, castability, or wear resistance is the priority, cast iron outperforms steel. The engineer chooses the material based on service requirements — a technician who understands why helps avoid misapplication.

5 Safety is not optional!

PPE is mandatory in the shop — no exceptions. We work with heat, sharp edges, and hazardous materials every lab session. Protect yourself and your classmates.

Resources & What's Next

Week 5 Resources

- Moniz Ch. 16 & 17 — Cast Irons & Stainless Steels
- Kalpakjian Ch. 6 — Ferrous Metals and Alloys

Visualizers

- Iron-Carbon Phase Diagram (Interactive)
- AISI/SAE Steel Designation Codes (Reference)

Coming Up — Week 6 Nonferrous Metals & Phase Diagrams

- Aluminum, copper, and titanium — why each family exists alongside steel
- Reading the iron-carbon phase diagram — phases, regions, and key transformation points
- From equilibrium to real-world cooling — how CCT diagrams show what actually forms
- Unit 2 Quiz covers Weeks 4–6 — review steel grades, heat treatment, and nonferrous metals